

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **G.GOWTHAMI(192524285)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. GOWTHAMI

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

ANSWERS

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Define the propositions:

F = Patient A has Fever

C = Patient A has Cough

R = Patient A has Rash

B = Patient A has Breathlessness

FL = Patient A has Possible Flu

M = Patient A has Possible Measles

P = Patient A has Possible Pneumonia

Rules:

Fever AND Cough $\rightarrow$ Possible Flu

$F \wedge C \rightarrow FL$

Fever AND Rash $\rightarrow$ Possible Measles

$F \wedge R \rightarrow M$

Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

$C \wedge B \rightarrow P$

Facts:

F = True, C = True, B = True

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules whose premises are satisfied.

Initial facts:

F = True

C = True

B = True

Step 1: Apply Rule 1

Rule:

$F \wedge C \rightarrow FL$

Since F and C are true:

$F \wedge C \Rightarrow FL$

Therefore:

Possible Flu = True

Step 2: Apply Rule 2

Rule:

$F \wedge R \rightarrow M$

We know F is true, but R is not given.

Therefore, Rule 2 cannot fire.

Step 3: Apply Rule 3

Rule:

$C \wedge B \rightarrow P$

Since C and B are both true:

$C \wedge B \Rightarrow P$

Therefore:

Possible Pneumonia = True

Final conclusions

Possible Flu = True

Possible Measles = Cannot be derived

Possible Pneumonia = True

Thus, the system derives that Patient A has Possible Flu and Possible Pneumonia.

(c) Backward Chaining for “Patient A has Pneumonia”

Goal:

P

Find a rule whose conclusion is P.

Rule 3:

$C \wedge B \rightarrow P$

Therefore, the goal P is reduced to two subgoals:

P

|— C (Cough) → Fact = True

|— B (Breathlessness) → Fact = True

Both subgoals are satisfied by the knowledge base.

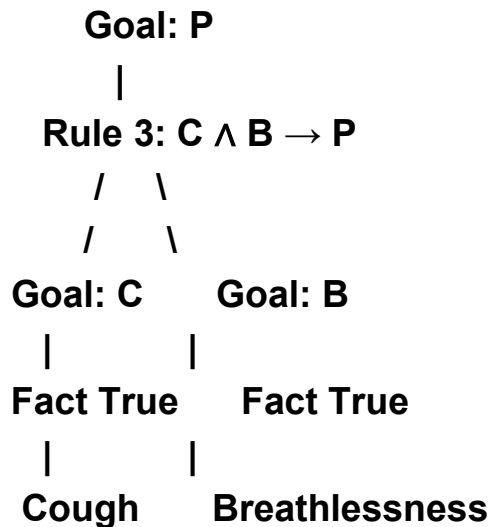
Therefore:

$C \wedge B \Rightarrow P$

Hence:

Patient A has Pneumonia = True

Goal Reduction Tree



Therefore, the goal Pneumonia is successfully proved.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification with Rule 1

Rule 1:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Given facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

The variables in Rule 1 are:

x

Match:

Vehicle(x) with Vehicle(Ambulance)

Therefore:

x = Ambulance

So,

$\theta_1 = \{x/\text{Ambulance}\}$

Next:

EmergencyType(x) with EmergencyType(Ambulance)

Again:

x = Ambulance

Thus:

$\theta_2 = \{x/\text{Ambulance}\}$

MGU

$\theta = \{x/\text{Ambulance}\}$

Applying this substitution to Rule 1:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$

Since both premises are facts:

$\text{ClearPath}(\text{Ambulance})$ is derived.

(b) Resolution Proof of $\text{Proceed}(\text{CarA})$

Relevant rules:

Rule 1

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This gives two clauses:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Resolution steps

From Rule 1 and the facts:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Resolve with:

$\text{Vehicle}(\text{Ambulance})$

and

EmergencyType(Ambulance)

giving:

ClearPath(Ambulance)

Now use Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Resolve with:

ClearPath(Ambulance)

giving:

GreenSignal(Ambulance)

Now use Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Substitute:

$x = \text{CarA}, y = \text{Ambulance}$

giving:

$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolve with:

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

Therefore:

Proceed(CarA)

Resolution chain

Vehicle(Ambulance)

+

EmergencyType(Ambulance)

↓

ClearPath(Ambulance)

↓

GreenSignal(Ambulance)

↓

Vehicle(CarA)

+

Behind(CarA,Ambulance)

+

GreenSignal(Ambulance)



Proceed(CarA)

Hence, **Proceed(CarA)** is logically entailed by the given KB.

(c) Two simultaneous emergency vehicles

The current knowledge base can handle individual emergency vehicles, because Rule 1 is universally quantified:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

For example, if another emergency vehicle **FireTruck** is added:

Vehicle(FireTruck)

EmergencyType(FireTruck)

then Rule 1 derives:

ClearPath(FireTruck)

Rule 2 then derives:

GreenSignal(FireTruck)

However, the KB does not adequately represent conflicts between two simultaneous emergency vehicles. It does not specify:

Which vehicle gets priority.

How two emergency routes are coordinated.

What happens when their paths intersect.

How conflicting green signals are handled.

Additional rules could include:

$\text{Priority}(\text{Ambulance}, \text{FireTruck}) \rightarrow \text{GivePriority}(\text{Ambulance})$

and:

$\text{GivePriority}(x) \rightarrow \text{GreenSignal}(x)$

A conflict rule could also be added:

$\text{Conflict}(x, y) \wedge \text{Priority}(x, y) \rightarrow \text{Hold}(y)$

Therefore, additional priority, conflict-detection, and route-coordination facts/rules are required for safe handling of simultaneous emergencies.

Case Study 3 — Agricultural Expert Reasoning System

(a) CNF Conversion

P1

Original:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Let:

$S = \text{SoilDry}$

I = IrrigationNeeded

W = CropWheat

D = ApplyDripMethod

R = CropAtRisk

Eliminate implication:

$\neg S \vee I$

So the CNF is:

P1: $\neg S \vee I$

P2

Original:

$IrrigationNeeded \wedge CropWheat \rightarrow ApplyDripMethod$

Eliminate implication:

$\neg(I \wedge W) \vee D$

Using De Morgan's law:

$\neg I \vee \neg W \vee D$

Therefore:

P2: $\neg I \vee \neg W \vee D$

This is already a CNF clause.

P3

Original:

$\neg ApplyDripMethod \wedge CropWheat \rightarrow CropAtRisk$

Eliminate implication:

$\neg(\neg D \wedge W) \vee R$

Using De Morgan's law:

$D \vee \neg W \vee R$

Therefore:

P3: $D \vee \neg W \vee R$

Facts

SoilDry = True

CNF:

S

CropWheat = True

CNF:

W

Complete CNF

$\neg S \vee I$

$\neg I \vee \neg W \vee D$

$D \vee \neg W \vee R$

S

W

(b) Resolution Refutation for ApplyDripMethod

Step 1: Goal

Goal:

D

For resolution refutation, negate the goal:

$\neg D$

Add it to the KB.

So we have:

$\neg S \vee I$

$\neg I \vee \neg W \vee D$

$D \vee \neg W \vee R$

S

W

$\neg D$

Step 2: Resolve

From:

$\neg S \vee I$

and:

S

resolve S:

I

So:

I

Step 3

Use:

$\neg I \vee \neg W \vee D$

with:

I

Resolve I:

$\neg W \vee D$

Step 4

Use:

W

with:

$\neg W \vee D$

Resolve W:

D

Step 5

We also have the negated goal:

$\neg D$

Resolve:

D

and

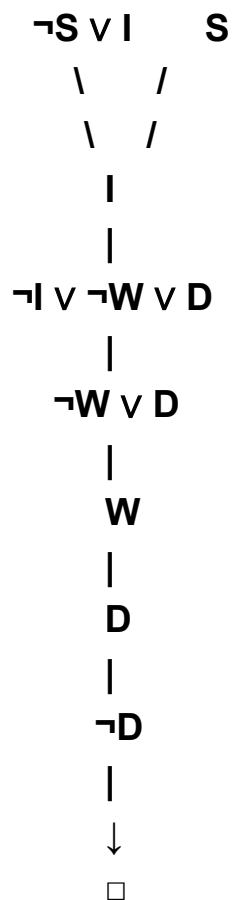
$\neg D$

Therefore:

□

The empty clause is obtained.

Resolution proof tree



Therefore:

$KB \models \text{ApplyDripMethod}$

Hence, ApplyDripMethod is logically proved.

(c) Removing SoilDry

If SoilDry (S) is removed, the fact S is no longer available.

The KB becomes:

$\neg S \vee I$

$\neg I \vee \neg W \vee D$

$D \vee \neg W \vee R$

W

Step 1

We cannot derive IrrigationNeeded (I) because the rule

$\neg S \vee I$

requires S to be true to infer I.

Since S is absent:

I cannot be derived.

Step 2

Without I, P2 cannot derive D:

$I \wedge W \rightarrow D$

Although W is true, I is unknown.

Therefore:

ApplyDripMethod cannot be proved.

Step 3: Does CropAtRisk follow?

Rule P3 is:

$\neg D \wedge W \rightarrow R$

or CNF:

$D \vee \neg W \vee R$

We know:

W = True

but we do not know:

$\neg D$

Therefore, CropAtRisk cannot be derived either.

Final result after removing SoilDry

Proposition

Result

SoilDry

Unknown

IrrigationNeeded

Cannot be derived

CropWheat

True

ApplyDripMethod

Cannot be derived

CropAtRisk

Cannot be derived

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Given percepts

The agent perceives:

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Let:

$S[1,2]$ = Stench at [1,2]

$B[1,1]$ = Breeze at [1,1]

$G[2,2]$ = Glitter at [2,2]

$W[x,y]$ = Wumpus at [x,y]

$P[x,y]$ = Pit at [x,y]

$Gold[x,y]$ = Gold at [x,y]

$Safe[x,y]$ = Cell [x,y] is safe

Initial belief state

$KB_0 = \{S[1,2], B[1,1], G[2,2]\}$

Modus Ponens — Rule 1

Rule:

$S[1,2] \rightarrow$ Wumpus is adjacent to [1,2]

Since:

$S[1,2] = \text{True}$

Therefore, by Modus Ponens:

Wumpus is adjacent to [1,2]

The adjacent cells of [1,2] are:

[1,1]

[1,3]

[2,2]

Therefore:

$W[1,1] \vee W[1,3] \vee W[2,2]$

This means the Wumpus is possibly in one of these cells; it does not identify one exact location.

Modus Ponens — Rule 2

Rule:

$B[1,1] \rightarrow \text{Pit is adjacent to } [1,1]$

Since:

$B[1,1] = \text{True}$

Therefore:

Pit is adjacent to [1,1]

The adjacent cells are:

[1,2]

[2,1]

Therefore:

$P[1,2] \vee P[2,1]$

Modus Ponens — Rule 3

Rule:

$G[x,y] \rightarrow \text{Gold}[x,y]$

Given:

$G[2,2] = \text{True}$

Therefore:

$\text{Gold}[2,2] = \text{True}$

Rule 4

Rule:

$\neg W[x,y] \wedge \neg P[x,y] \rightarrow \text{Safe}[x,y]$

To apply this rule, the agent must know both:

No Wumpus exists in the cell.

No Pit exists in the cell.

The current percepts do not establish these negative facts for any particular cell.

Therefore, Rule 4 cannot currently derive a new safe cell.

Derived conclusions

Percept/Inference

Conclusion

Stench[1,2]

Wumpus is adjacent to [1,2]

Breeze[1,1]

Pit is adjacent to [1,1]

Glitter[2,2]

Gold is at [2,2]

Rule 4

No new Safe fact can currently be derived

Thus the belief state contains:

{Stench[1,2], Breeze[1,1], Glitter[2,2], Wumpus adjacent to [1,2], Pit adjacent to [1,1], Gold[2,2]}

(b) Forward Chaining — Possible Wumpus and Pit Locations

Step 1: Wumpus

Given:

Stench[1,2]

Rule:

Stench[1,2] → Wumpus adjacent to [1,2]

The adjacent cells are:

[1,1], [1,3], [2,2]

Therefore:

Possible Wumpus locations = {[1,1], [1,3], [2,2]}

So the agent cannot conclude that the Wumpus is at one specific cell.

Step 2: Pit

Given:

Breeze[1,1]

Rule:

Breeze[1,1] → Pit adjacent to [1,1]

Adjacent cells:

[1,2], [2,1]

Therefore:

Possible Pit locations = {[1,2], [2,1]}

Again, the agent cannot identify the exact pit location from this single percept.

Step 3: Gold

Given:

Glitter[2,2]

Rule:

Glitter[x,y] → Gold[x,y]

Therefore:

Gold[2,2]

Forward-chaining summary

Stench[1,2]



Wumpus adjacent to [1,2]



Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]



Pit adjacent to [1,1]



Possible Pit:

[1,2], [2,1]

Glitter[2,2]



Gold[2,2]

Therefore:

Wumpus $\in \{[1,1], [1,3], [2,2]\}$

Pit $\in \{[1,2], [2,1]\}$

Gold = [2,2]

(c) Safety of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The proposed path is:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

We must check each cell.

Cell [1,1]

The agent has a Breeze at [1,1].

This means:

$\text{Pit}[1,2] \vee \text{Pit}[2,1]$

Thus, the percept at [1,1] indicates that an adjacent pit exists, but it does not prove that [1,1] itself contains a pit.

So [1,1] itself is not identified as dangerous by the given rules.

Cell [1,2]

This cell has a Stench.

More importantly, from the Breeze at [1,1]:

$\text{Pit}[1,2] \vee \text{Pit}[2,1]$

Therefore:

[1,2] is a possible Pit location.

Also, the Stench at [1,2] indicates that a Wumpus is adjacent to [1,2]:

$\text{Wumpus}[1,1] \vee \text{Wumpus}[1,3] \vee \text{Wumpus}[2,2]$

Therefore, [1,2] cannot be confirmed safe.

Cell [2,2]

We know:

$\text{Gold}[2,2]$

However, the Stench at [1,2] gives:

$\text{Wumpus}[1,1] \vee \text{Wumpus}[1,3] \vee \text{Wumpus}[2,2]$

Therefore, [2,2] is also a possible Wumpus location.

The given KB does not provide enough negative information to establish:

$\neg \text{Wumpus}[2,2] \wedge \neg \text{Pit}[2,2]$

Hence Rule 4 cannot prove:

$\text{Safe}[2,2]$

Final conclusion

The complete path

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

cannot be guaranteed safe using the current knowledge base.

In particular:

[1,2] is a possible Pit location.

[2,2] is a possible Wumpus location.

[2,2] contains the gold, but the agent has not logically established that it is safe.

Therefore, the rational logical-agent conclusion is:

Do not take the complete path yet. The current KB is insufficient to prove the path safe.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand